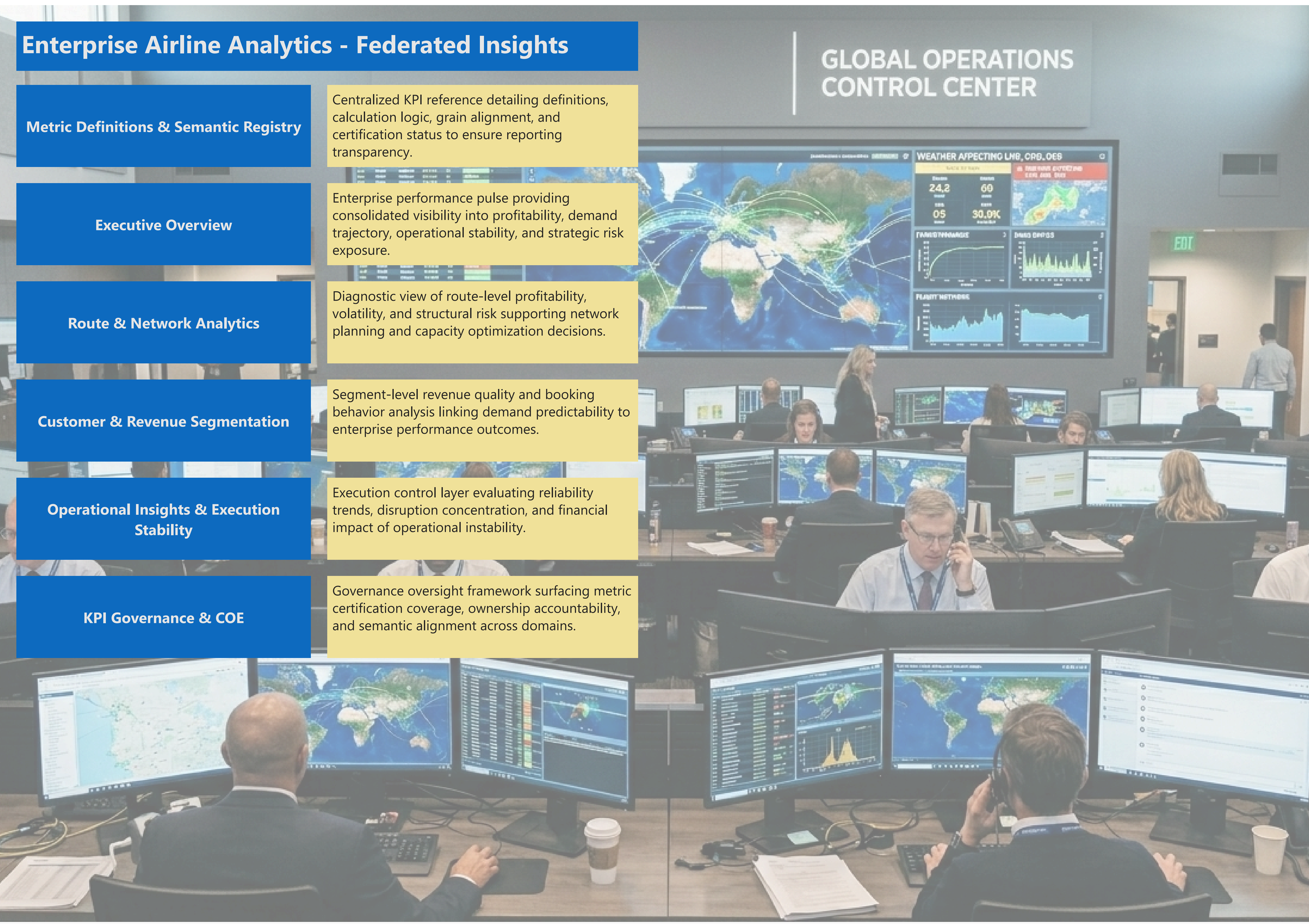
Enterprise Airline Analytics - Federated Insights

Metric Definitions & Semantic Registry	Centralized KPI reference detailing definitions, calculation logic, grain alignment, and certification status to ensure reporting transparency.
Executive Overview	Enterprise performance pulse providing consolidated visibility into profitability, demand trajectory, operational stability, and strategic risk exposure.
Route & Network Analytics	Diagnostic view of route-level profitability, volatility, and structural risk supporting network planning and capacity optimization decisions.
Customer & Revenue Segmentation	Segment-level revenue quality and booking behavior analysis linking demand predictability to enterprise performance outcomes.
Operational Insights & Execution Stability	Execution control layer evaluating reliability trends, disruption concentration, and financial impact of operational instability.
KPI Governance & COE	Governance oversight framework surfacing metric certification coverage, ownership accountability, and semantic alignment across domains.



METRIC DEFINITIONS & SEMANTIC REGISTRY

KPI	Category	Business Definition	Calculation Logic	Source	Function	Grain	Certified	Status	Version
% Certified KPIs	Governance	Percentage of KPIs certified	Certified KPI Count / Total KPIs	Dim_KPI_Definitions	CoE	Metadata	Yes	Certified	1.00
Advance Booking Window (Days)	Transformational	Average number of days between booking and flight date	AVERAGE(Booking_Lead_Days)	Fact_Bookings	Commercial	Daily	Yes	Certified	1.00
Ancillary Revenue %	Commercial	Percentage of revenue from ancillary services	Ancillary Revenue / Total Revenue	Fact_Bookings	Commercial	Daily	Yes	Certified	1.00
Cancellation Rate %	Operational	Cancelled flights divided by total flights	Cancelled Flights / Total Flights	Fact_Flights	Operations	Daily	Yes	Certified	1.00
Certified KPI Count	Governance	Total number of KPIs marked as certified	COUNT where Certified_Flag = Yes	Dim_KPI_Definitions	CoE	Metadata	Yes	Certified	1.00
Cost per Route	Financial	Average monthly operating cost per route	Total_Cost by Route	Fact_Financials	Finance	Monthly	Yes	Certified	1.00
Demand Momentum Index	Transformational	3-month rolling growth rate in tickets sold	Rolling 3M Tickets vs Previous 3M Tickets	Fact_Bookings	Commercial	Monthly	Yes	Certified	1.00
Fuel Cost Ratio	Financial	Fuel cost as percentage of total operating cost	Fuel_Cost / Total_Cost	Fact_Financials	Finance	Monthly	Yes	Certified	1.00
Load Factor %	Operational	Seats sold divided by total seat capacity	Seats_Sold / Seats_Capacity	Fact_Flights	Operations	Daily	Yes	Certified	1.00
On-Time Departure %	Operational	Flights departing within 15 minutes of schedule	On-Time Flights / Total Flights	Fact_Flights	Operations	Daily	Yes	Certified	1.00
Premium Cabin Revenue %	Commercial	Revenue share from premium cabin classes	Premium Revenue / Total Revenue	Fact_Bookings	Commercial	Daily	Yes	Certified	1.00
Profit Margin %	Executive	Route profit divided by total revenue	Route Profit / Total Revenue	Composite	Finance	Monthly	Yes	Certified	1.00
Revenue Forecast Accuracy %	Transformational	Accuracy of revenue forecast relative to actuals	1 - ABS((Actual Revenue - Forecast Revenue) / Forecast Revenue)	Fact_Bookings	CoE	Monthly	Yes	Certified	1.00
Revenue per Flight	Commercial	Average revenue generated per flight	Total Revenue / Distinct Flights	Composite	Commercial	Daily	Yes	Certified	1.00
Route Profit	Executive	Revenue minus total operating cost	[Total Revenue] - [Total Operating Cost]	Composite	Finance	Monthly	Yes	Certified	1.00
Route Risk Score	Transformational	Composite score combining load volatility, margin volatility and cancellation rate	Standardized composite risk index	Composite	CoE	Monthly	No	Draft	1.00
Ticket Forecast Accuracy %	Transformational	Accuracy of ticket forecast relative to actuals	1 - ABS((Actual Tickets - Forecast Tickets) / Forecast Tickets)	Fact_Bookings	CoE	Monthly	Yes	Certified	1.00
Total Operating Cost	Executive	Total operational cost including fuel, staff and airport fees	SUM(Fact_Financials[Total_Cost])	Fact_Financials	Finance	Monthly	Yes	Certified	1.00
Total Passengers	Executive	Total tickets sold across routes	SUM(Fact_Bookings[Tickets_Sold])	Fact_Bookings	Commercial	Daily	Yes	Certified	1.00
Total Revenue	Executive	Total passenger and ancillary revenue generated	SUM(Fact_Bookings[Total_Revenue])	Fact_Bookings	Commercial	Daily	Yes	Certified	1.00
Yield per Passenger	Commercial	Revenue per ticket sold	Total Revenue / Total Passengers	Fact_Bookings	Commercial	Daily	Yes	Certified	1.00

For questions & inquiries, contact Anil Jacob at Aniljacobs@gmail.com, or visit my GitHub
This project repo is available at the following link: https://github.com/AwesomeAnil/Federated-Insights-Case-Study

EXECUTIVE OVERVIEW - YEAR 2024

Close Panel

Hide Yellow box commentary

Show Yellow box commentary

Select a Year:

- ○ 2022
- ○ 2023
- ◉ 2024

Select a route:

- ☐ DXB-BOM
- ☐ DXB-JFK
- ☐ DXB-LHR
- ☐ DXB-SIN
- ☐ DXB-SYD

Select a fare class:

- ☐ Economy
- ☐ Premium Economy
- ☐ Business
- ☐ First

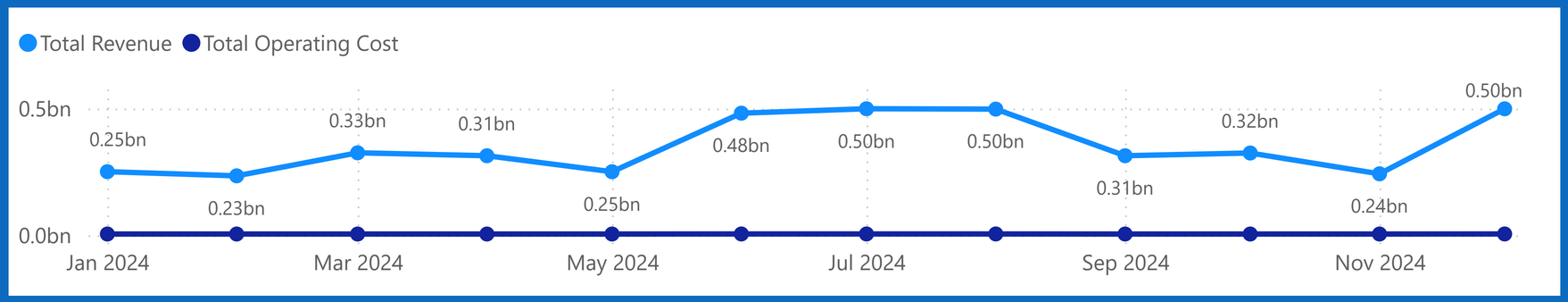
Select a segment:

- ☐ Business Traveller
- ☐ Corporate Contract
- ☐ Leisure Traveller
- ☐ Loyalty Gold

Executive KPIs

4.2bn	4.2bn	98.6%	89.7%	-0.13
Total Revenue	Route Profit	Profit Margin %	Revenue Forecast A...	Demand Momentu...

Financial Performance & Trends



Demand & Operational Signals

79.9%	39.3%
Load Factor %	On-Time Departure %
4.4%	**71.83**
Cancellation Rate %	Average of Tickets S...

Risk & Exposure

Route_ID	Route Risk Score	Profit Margin %	Load Factor %
DXB-SYD	2.69	98.3%	79.9%
DXB-JFK	2.64	98.5%	80.0%
DXB-LHR	2.62	98.7%	79.8%
DXB-BOM	2.61	98.8%	80.0%
Total	**2.63**	**98.6%**	**79.9%**

0.44	2.63
Fuel Cost Ratio	Route Risk Score

Executive KPI's

Consolidated view of revenue growth, profitability, forecast reliability, and forward demand momentum to assess overall business health and trajectory.

Financial Performance Trends:

Illustrates revenue progression, operating cost pressure, and margin resilience across seasonal demand cycles and fuel volatility impacts.

Demand and Operational Signals:

Surfaces booking acceleration trends and operational stability indicators that influence near-term revenue realization and customer reliability.

Risk & Exposure:

Highlights route-level volatility, margin sensitivity, and operational disruption risk to support proactive network and cost management decisions.

ROUTE ANALYTICS (NETWORK PLANNING) - YEAR 2024

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Show Yellow box commentary

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- ○ 2023
- ◉ 2024

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- ☐ First

Select a segment:

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- ☐ Corporate Contract
- ☐ Leisure Traveller
- ☐ Loyalty Gold

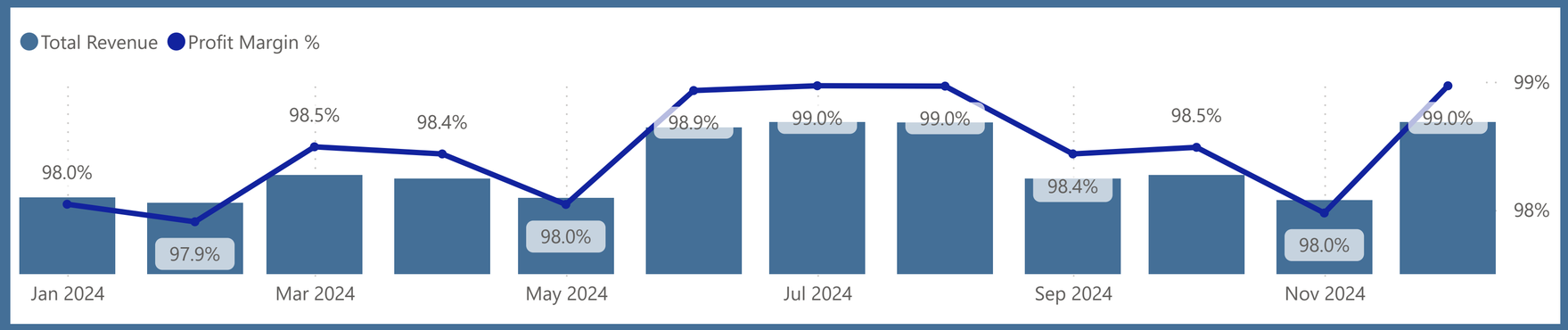
Route Health Snapshot

4.2bn	98.6%	79.9%	2.63	4.99M
Total Revenue	Profit Margin %	Load Factor %	Route Risk Score	Cost per Route

Route Health Snapshot

Provides a consolidated health view of the selected route, combining profitability, demand strength, and structural risk to assess strategic viability.

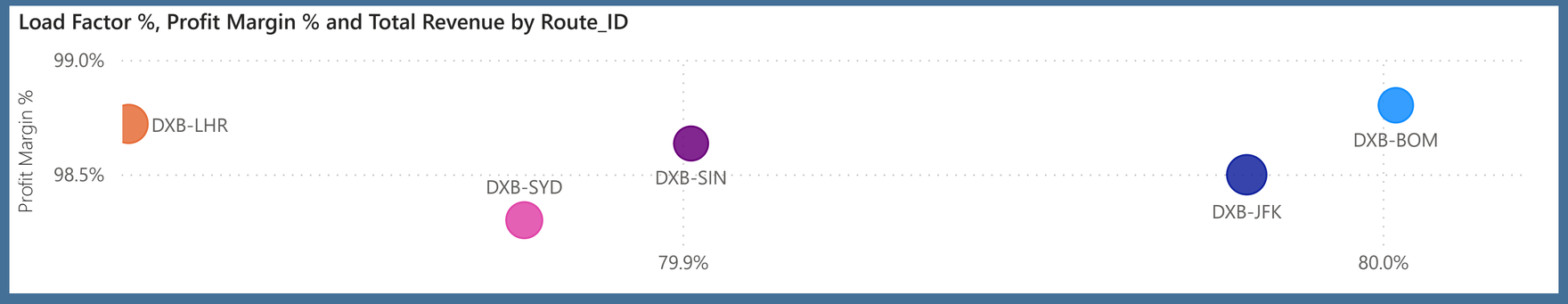
Route Revenue & Margin Trend



Route Revenue & Margin Trend

Reveals whether profitability is improving, deteriorating, or structurally constrained by cost pressure.

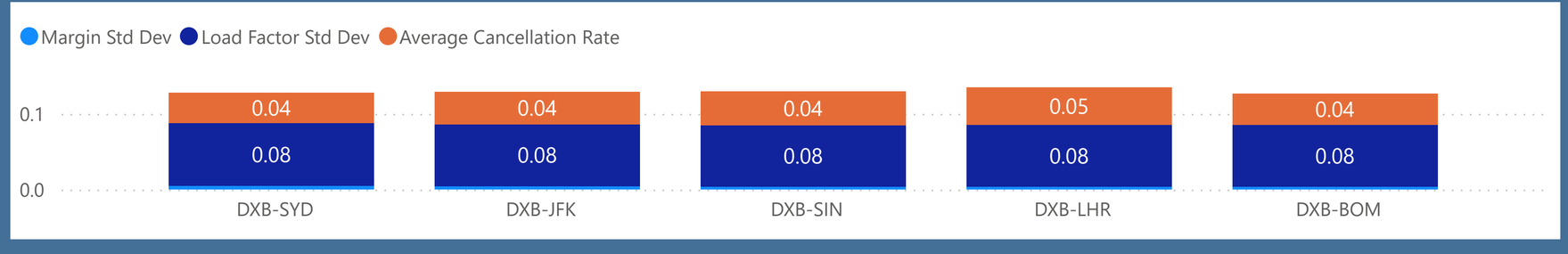
Performance Quadrant



Performance Quadrant

Categorizes network performance by load strength and margin resilience to guide capacity and pricing decisions.

Risk & Volatility breakdown



Risk & Volatility

Deconstructs the volatility drivers influencing route-level risk exposure across margin stability, demand variability, and operational disruption.

CUSTOMER & COMMERCIAL SEGMENTATION- YEAR 2024

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Show Yellow box commentary

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- ○ 2023
- ◉ 2024

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- ☐ DXB-SIN
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- ☐ First

Select a segment:

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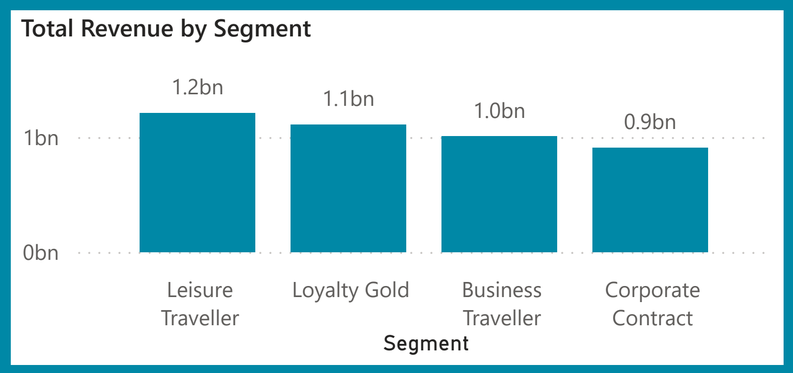
- ☐ Corporate Contract
- ☐ Leisure Traveller
- ☐ Loyalty Gold

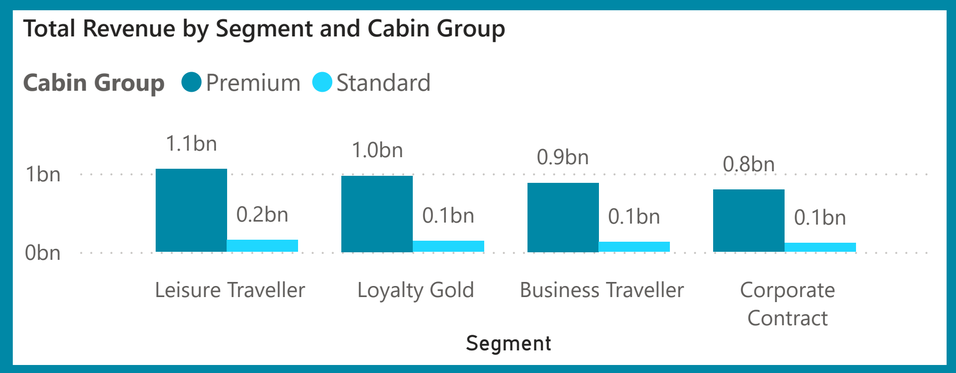
Segment Performance Snapshot

4.2bn	2,013	87.3%	2M	45
Total Revenue	Yield per Passenger	Premium Cabin Reve...	Total Passengers	Advance Booking Wi...

Segment Performance Snapshot:
Consolidates segment-level revenue contribution, yield quality, and booking behavior to assess structural demand value.

Revenue & Mix Analysis

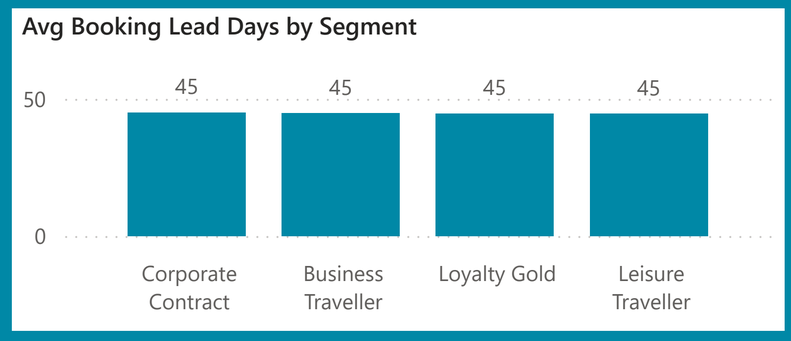


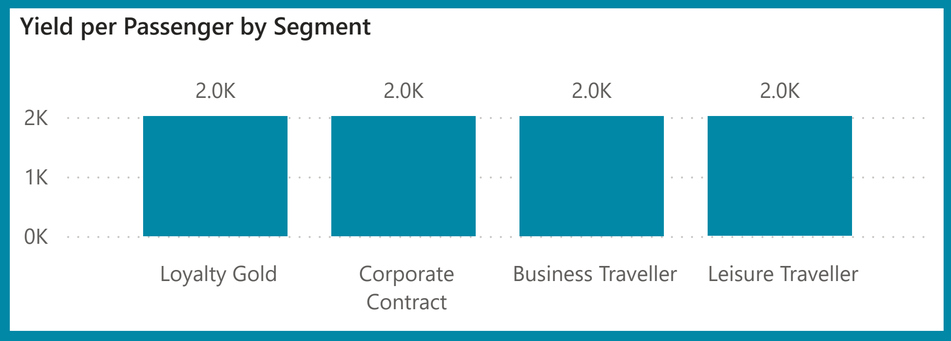


Revenue & Segment Mix Analysis:

Evaluates revenue concentration and premium mix distribution to highlight structurally higher-margin customer groups.

Booking Behavior & Yield Analysis

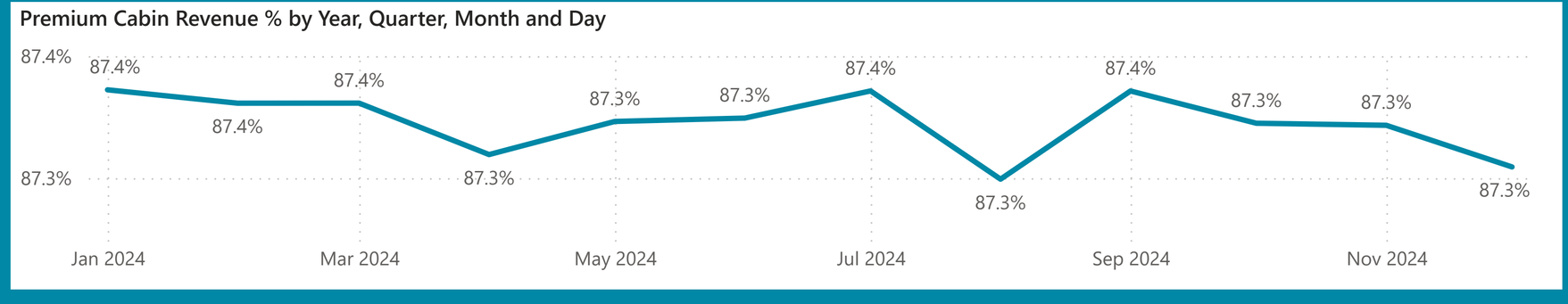




Booking Lead Days:

Reveals advance booking behavior differences across segments to assess demand visibility and forecasting reliability.

Premium Demand Trend



Premium Booking Trend:

Tracks evolution of premium revenue share across segments to detect long-term shifts in customer mix and profitability drivers.

OPERATIONAL INSIGHT (EXEC CONTROL VIEW) - YEAR 2022

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Select a Year:

- (Blank)
- 2022
- 2023
- 2024

Select a route:

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- DXB-JFK
- DXB-LHR
- DXB-SIN
- DXB-SYD

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- Premium Economy
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- First

Select a segment:

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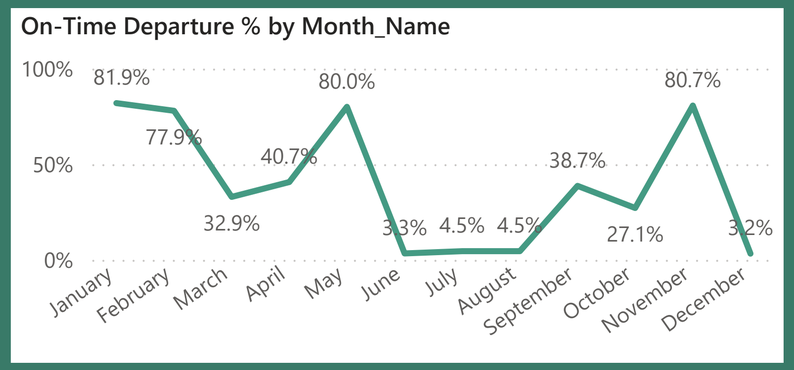
- Corporate Contract
- Leisure Traveller
- Loyalty Gold

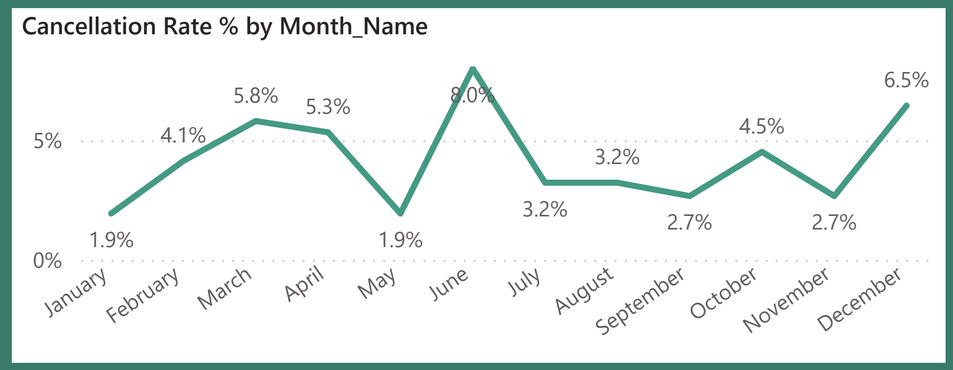
Operational Stability Snapshot

39.4%	4.2%	79.8%	2M	4.2bn
On-Time Departure %	Cancellation Rate %	Load Factor %	Total Passengers	Total Revenue

Operational Stability Snapshot:
Provides a consolidated view of execution reliability and disruption exposure to assess overall operational stability

Delay & Cancellation Trends

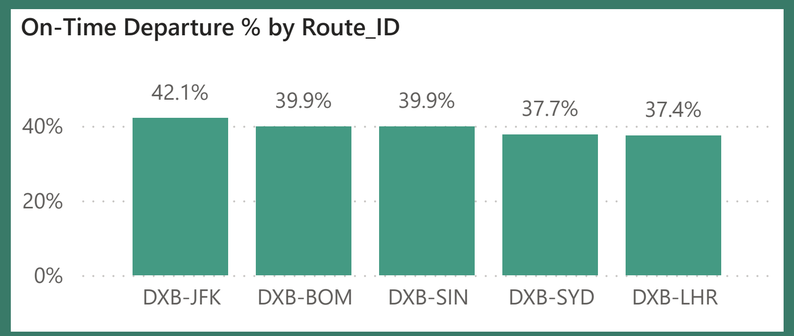


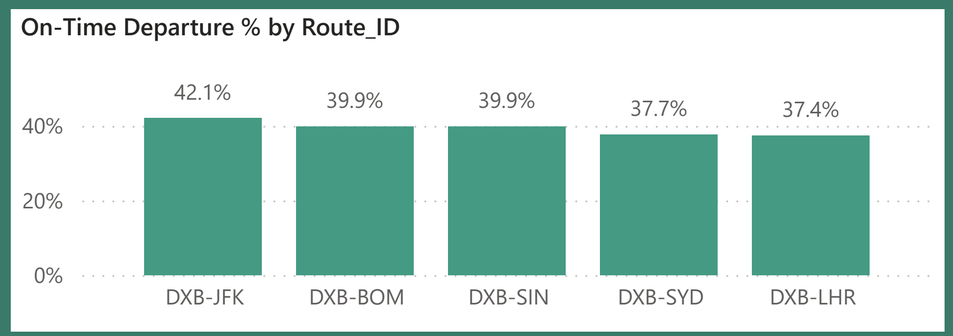


Delay & Cancellation Trends:

Reveals structural reliability patterns over time to distinguish temporary volatility from sustained operational weakness.

Operational Concetration

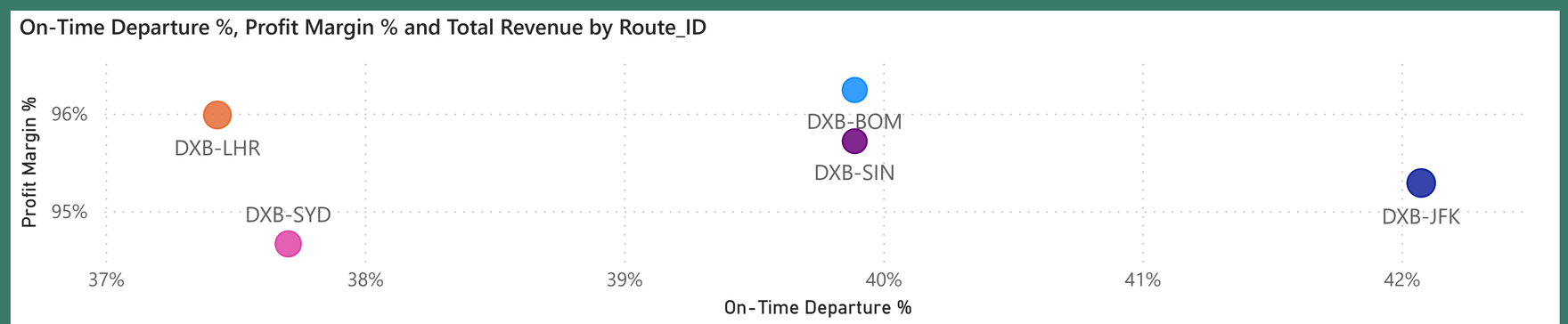




Operational Concentration:

Identifies routes or periods exhibiting disproportionate disruption, enabling targeted intervention rather than broad corrective measures.

Margin Impact Linkage



Linking Margin Impact:

Evaluates whether operational instability is materially compressing route-level profitability and elevating enterprise risk exposure.

KPI GOVERNANCE & ANALYTICS CoE

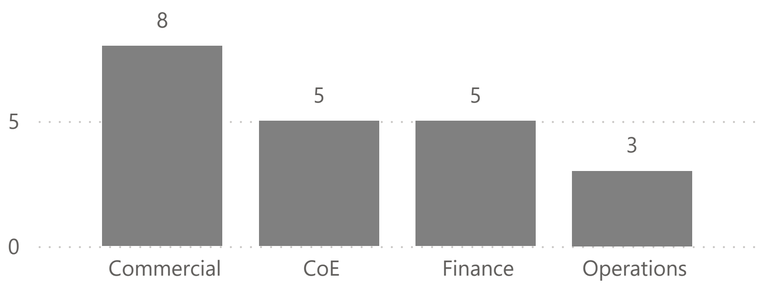
Governance Coverage Snapshot

21	20	95.2%	1	4
Total KPI's	Certified KPI's	Certified Coverage%	Draft KPI's	Domains

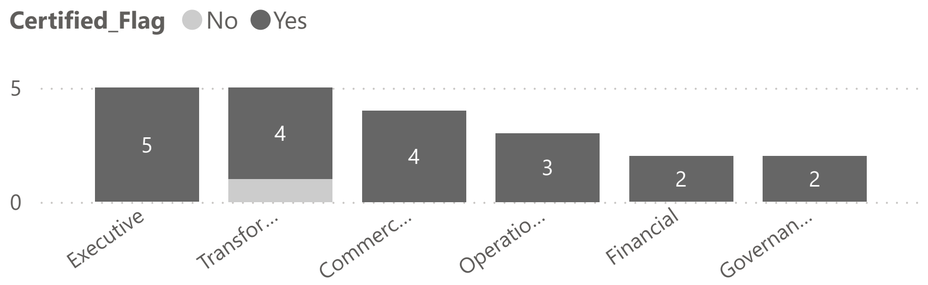
Governance & Coverage Snapshot:
Provides executive visibility into governance maturity and metric standardization across the analytics ecosystem.

KPI Domain Distribution

Total KPI's by Domain



Total KPI's by KPI_Category and Certified_Flag



KPI Domain Distribution:

Evaluates KPI distribution across enterprise domains to ensure balanced analytical coverage and standardization.

Ownership & Accountability Matrix

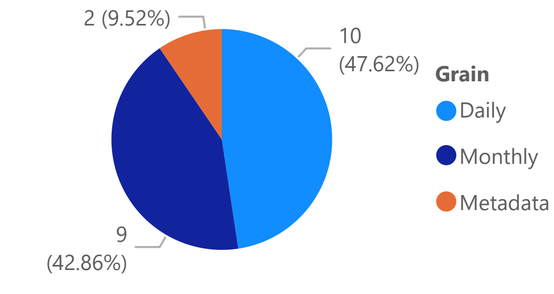
KPI_Name	Owner_Function	Certification_Status	Grain	Fact_Source	Certified_Flag	Calculation_Logic
% Certified KPIs	CoE	Certified	Metadata	Dim_KPI_Definitions	Yes	Certified KPI Count / Total KPIs
Advance Booking Window (Days)	Commercial	Certified	Daily	Fact_Bookings	Yes	AVERAGE(Booking_Lead_Days)
Ancillary Revenue %	Commercial	Certified	Daily	Fact_Bookings	Yes	Ancillary Revenue / Total Revenue
Cancellation Rate %	Operations	Certified	Daily	Fact_Flights	Yes	Cancelled Flights / Total Flights
Certified KPI Count	CoE	Certified	Metadata	Dim_KPI_Definitions	Yes	COUNT where Certified_Flag = Yes
Cost per Route	Finance	Certified	Monthly	Fact_Financials	Yes	Total_Cost by Route

Ownership & Accountability Matrix:

Surfaces metric ownership, certification status, and calculation grain to enforce accountability and semantic consistency.

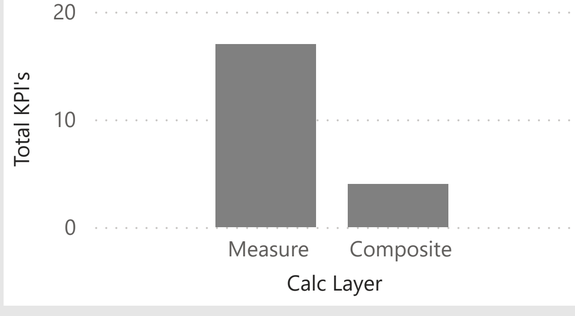
Registry Integrity & Grain Alignment

Total KPI's by Grain



KPI_Name	Grain
Advance Booking Window (Days)	Daily
Ancillary Revenue %	Daily
Cancellation Rate %	Daily
Load Factor %	Daily
On-Time Departure %	Daily
Premium Cabin Revenue %	Daily
Revenue per Flight	Daily

Total KPI's by Calc Layer



Registry Integrity & Grain Alignment:

Validates metric grain alignment and registry integrity to prevent cross-grain ambiguity and inconsistent reporting.